

IOWA CITY, IA, USA

WMO: 725462

Lat: 41.633N

Lon: 91.543W

Elev: 198

StdP: 98.97

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 14937

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.3	-17.4	-24.3	0.4	-19.9	-22.1	0.5	-16.9	11.9	-3.7	11.0	-5.2	3.3	310	0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	33.3	24.5	32.0	24.1	30.4	23.1	26.4	30.9	25.4	30.0	24.4	28.9	4.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	20.7	29.5	23.9	19.2	28.3	22.9	18.0	27.1	83.4	31.0	78.7	29.9	74.7	28.9	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	9.0	8.0	-25.0	35.5	3.4	1.9	-27.4	36.9	-29.4	38.1	-31.3	39.2	-33.8	40.6		
(5)			-25.2	28.2	3.3	0.7	-27.6	28.6	-29.5	29.0	-31.4	29.4	-33.8	29.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.3	-5.1	-3.3	3.6	10.6	16.7	21.9	24.0	22.6	18.8	11.5	4.4	-2.6		
	DBStd	11.47	6.61	6.67	6.32	5.21	4.66	3.42	3.17	2.85	4.29	5.22	5.55	6.39		
	HDD10.0	1733	469	373	214	54	4	0	0	0	1	44	181	393		
	HDD18.3	3458	727	605	456	237	88	9	1	3	46	219	417	649		
	CDD10.0	1858	0	1	18	71	212	358	433	392	266	91	14	2		
	CDD18.3	541	0	0	1	4	37	117	176	137	61	8	0	0		
	CDH23.3	5027	0	0	8	68	351	1061	1732	1136	581	91	1	0		
	CDH26.7	1731	0	0	1	13	91	366	673	372	196	21	0	0		
(14) Wind	WSAvg	3.5	3.9	3.9	4.1	4.2	3.7	3.1	2.7	2.5	2.7	3.3	3.7	3.6		
(15) Precipitation	PrecAvg	945	28	34	57	96	119	146	115	109	75	68	57	40		
	PrecMax	1521	54	99	136	211	315	289	350	295	179	189	176	97		
	PrecMin	592	9	10	12	37	16	37	16	6	24	13	4	12		
	PrecStd	206	14	21	35	43	59	69	75	68	40	44	42	22		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.5	16.8	24.2	28.7	32.1	33.9	36.3	34.5	33.3	29.7	22.4	16.2		
		MCWB	7.5	11.4	17.1	18.3	22.6	24.4	25.8	24.0	23.1	20.9	14.8	13.4		
	2%	DB	8.1	12.1	20.3	25.3	28.9	32.3	33.7	32.1	31.2	26.1	18.6	12.0		
		MCWB	5.3	8.1	13.9	16.5	20.5	23.7	25.7	24.5	22.6	18.6	13.3	9.9		
	5%	DB	5.4	8.3	17.0	22.6	27.2	30.6	32.1	30.2	28.9	23.0	15.9	8.7		
		MCWB	3.2	5.8	12.6	14.8	19.4	22.8	24.9	23.5	21.7	17.1	11.4	6.6		
	10%	DB	3.0	5.7	13.3	19.9	25.1	28.8	30.3	28.6	27.0	20.5	13.0	5.9		
		MCWB	1.6	3.6	9.1	13.2	18.4	21.9	23.8	22.6	20.0	15.5	9.4	3.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.6	12.5	17.9	19.5	23.8	26.7	28.0	27.3	25.3	22.2	16.5	14.5		
		MCDB	10.5	15.9	23.2	26.0	29.6	31.1	33.0	31.7	30.7	27.6	20.4	16.1		
	2%	WB	5.6	8.7	15.4	17.6	22.1	25.4	26.8	25.7	23.9	20.1	14.1	10.0		
		MCDB	8.2	11.6	18.7	22.8	27.4	30.2	31.6	29.8	29.0	23.9	17.7	11.4		
	5%	WB	3.4	5.8	12.8	16.0	20.8	24.2	25.7	24.7	22.7	18.3	12.2	6.6		
		MCDB	5.2	8.4	16.5	21.4	25.2	29.0	30.4	28.7	26.9	21.5	15.4	8.3		
	10%	WB	1.8	3.5	9.5	14.2	19.5	23.1	24.7	23.7	21.4	16.3	9.7	3.8		
		MCDB	3.1	5.3	13.3	18.8	23.7	27.2	29.3	27.4	25.4	20.1	12.8	5.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.7	9.2	10.3	11.7	11.3	10.6	10.4	10.6	12.4	11.5	10.1	8.5		
		MCDBR	11.9	13.1	15.2	16.3	13.4	12.4	11.8	12.3	14.0	15.0	14.1	12.1		
	5% WB	MCWBR	9.3	9.4	9.5	9.1	6.9	5.9	5.3	5.9	6.6	8.2	9.4	9.6		
		MCWBR	10.7	12.5	13.3	13.7	11.8	10.8	10.6	10.5	11.7	12.1	13.1	11.0		
(40) Clear-Sky Solar Irradiance	taub		0.290	0.306	0.338	0.384	0.425	0.426	0.442	0.422	0.391	0.343	0.302	0.290		
	taud		2.404	2.373	2.378	2.264	2.213	2.263	2.204	2.267	2.316	2.450	2.491	2.429		
	Ebn at Noon		862	903	909	885	853	851	832	838	840	849	845	830		
	Edh at Noon		79	96	108	130	141	134	141	128	114	89	73	72		
(44) All-Sky Solar Radiation	RadAvg		1.82	2.81	3.77	4.71	5.54	6.16	6.17	5.42	4.58	3.02	2.03	1.48		
	RadStd		0.21	0.24	0.33	0.38	0.39	0.59	0.43	0.34	0.36	0.34	0.17	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page